

Representations of Lie Groups

A Mathematical Dissection of Quantum Spin

Part I: Mathematical Foundations

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Topological Space

A **topological space** is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T} \subseteq \mathcal{P}(X)$ satisfies:

- ① (Trivial Bounds) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
- ② (Arbitrary Unions) $\forall \{U_\alpha\}_{\alpha \in \mathcal{I}} \subseteq \mathcal{T} : \bigcup_{\alpha \in \mathcal{I}} U_\alpha \in \mathcal{T}$.
- ③ (Finite Intersections) $\forall U, V \in \mathcal{T} : U \cap V \in \mathcal{T}$.

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Key constructions: Product topology, subspace topology, metric topologies (e.g. Euclidean topology on $\mathbb{R}^n$).

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- ② (Second-Countable) The topology has a countable basis.
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Intuition: A manifold is a space that *locally looks like* $\mathbb{R}^n$, but may have a nontrivial global shape (e.g. a sphere S^2 or a torus T^2).

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Why Hausdorff + second-countable? These are technical conditions that prevent pathological spaces. They ensure partitions of unity exist.

Smooth Manifold

A **smooth manifold** is a triple $(M, \mathcal{T}, \mathcal{A})$ where:

- ① $(M, \mathcal{T})$ is a topological manifold.
- ② $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{I}}$ is a **smooth atlas**: each $(U_\alpha, \varphi_\alpha)$ is a chart, the charts cover M , and transition maps $\varphi_\beta \circ \varphi_\alpha^{-1}$ are C^∞ .
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Key idea: Charts are local coordinate systems. Smoothness means that when you switch between coordinate systems, the change-of-coordinates map is infinitely differentiable.

Smooth Maps and Diffeomorphisms

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A map $F : M \rightarrow N$ between smooth manifolds is **smooth** if its coordinate representation $\psi \circ F \circ \varphi^{-1}$ is C^∞ for all compatible charts.

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Examples: $\mathbb{R}^n$ is diffeomorphic to the open unit ball $B_1(0)$. S^n is NOT diffeomorphic to $\mathbb{R}^n$ (compact vs. non-compact).

Tangent Spaces via Derivations

Motivation

How do we define tangent vectors to a smooth manifold without embedding it in $\mathbb{R}^N$? Answer: **derivations** — linear maps that satisfy the Leibniz rule.

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Let M be a smooth manifold, $p \in M$. The **stalk** $C_p^\infty M$ is the set of germs of smooth functions at p . The **tangent space** at p is

$$T_p M \stackrel{\text{df}}{=} \{v : C_p^\infty M \rightarrow \mathbb{R} : | \, v \text{ is a derivation at } p\}.$$

$T_p M$ is an $\mathbb{R}$ -vector space with $\dim(T_p M) = \dim(M)$.

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Why derivations? For $\mathbb{R}^n$, a tangent vector v acts on functions by the directional derivative: $v(f) = \sum_i v^i \frac{\partial f}{\partial x^i} \Big|_p$. This algebraic characterization generalizes to any manifold.

The Differential

Differential

Let $F : M \rightarrow N$ be smooth, $p \in M$. The **differential of F at p** is the linear map $\mathbf{d}F_p : T_p M \rightarrow T_{F(p)} N$ defined by

$$\forall v \in T_p M, \forall [f]_{F(p)} \in C_{F(p)}^\infty N : \quad [(\mathbf{d}F_p(v))]([f]_{F(p)}) \stackrel{\text{df}}{=} v([f \circ F]_p).$$

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Interpretation: The differential is the “best linear approximation” to a smooth map, generalizing the Jacobian matrix from multivariable calculus.

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Why these matter: Lie groups combine the smooth structure of manifolds with the algebraic structure of groups. Lie algebras are vector spaces with an extra bracket operation.

$\mathbb{F}$ -Algebra

A vector space $\mathcal{A}$ over $\mathbb{F}$ together with a bilinear multiplication $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$. (Examples: $\mathbb{R}^{n,n}$ with matrix multiplication, $C^\infty(M)$ with pointwise multiplication.)

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An $\mathbb{F}$ -vector space $\mathfrak{g}$ with a bilinear bracket $[\bullet, \bullet] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying:

- ① (Skew-symmetry) $[Y, X] = -[X, Y]$
- ② (Jacobi identity) $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$

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Key example: $(\mathbb{R}^3, \times)$ — the cross product makes $\mathbb{R}^3$ a Lie algebra!
 $[v, w] = v \times w$ satisfies skew-symmetry and Jacobi.

Lie Groups

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Examples:

- $(\mathbb{R}^n, +)$ — the translation group.
- $\mathrm{GL}(n, \mathbb{R})$ — invertible $n \times n$ real matrices.
- $\mathrm{SO}(n)$ — rotations in $\mathbb{R}^n$.
- $\mathrm{SU}(n)$ — unitary transformations with determinant 1.

Left Translation and Left-Invariant Vector Fields

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Geometric meaning: A left-invariant vector field is completely determined by its value at the identity e . Translating the vector at e by L_g gives the vector at any point g .

The Lie Algebra of a Lie Group

Tangent Space at Identity is a Lie Algebra

Let G be a Lie group. Define $\mathfrak{g} = T_e G$ (tangent space at identity). For each $v \in \mathfrak{g}$, there exists a **unique** left-invariant vector field X^v with $X^v|_e = v$. Define the bracket on $\mathfrak{g}$ by

$$[v, w]_{\mathfrak{g}} \stackrel{\text{df}}{=} [X^v, X^w]_o|_e.$$

Then $(\mathfrak{g}, [\bullet, \bullet]_{\mathfrak{g}})$ is a Lie $\mathbb{R}$ -algebra with $\dim(\mathfrak{g}) = \dim(G)$.

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The bridge: Every Lie group gives rise to a Lie algebra (its tangent space at the identity). The Lie algebra captures the **infinitesimal** structure of the group.

The Closed Subgroup Theorem (É. Cartan)

Closed Subgroup Theorem

Let G be a Lie group and $H \subseteq G$ a closed subgroup. Then H admits a unique smooth structure making it a Lie group such that the inclusion $H \hookrightarrow G$ is a smooth embedding.

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Examples: $\mathrm{SO}(n)$, $\mathrm{SU}(n)$, $\mathrm{SL}(n, \mathbb{R})$, $\mathrm{Sp}(2n)$ are all closed subgroups of $\mathrm{GL}(N, \mathbb{F})$ and therefore Lie groups.

The General Linear Group $\text{GL}(n, \mathbb{F})$

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- Its Lie algebra is $\mathfrak{gl}(n, \mathbb{F}) = \mathbb{F}^{n,n}$ (all $n \times n$ matrices) with the commutator bracket:

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- Most matrix Lie groups arise as closed subgroups of $\mathrm{GL}(n, \mathbb{F})$. Their Lie algebras are subspaces of $\mathfrak{gl}(n, \mathbb{F})$ closed under the commutator.

The Special Orthogonal Group $\text{SO}(3)$

The Special Orthogonal Group (Rigid Rotations)

$$\text{SO}(3) = \{ \mathbf{R} \in \mathbb{R}^{3,3} : \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1 \};$$

$$\mathfrak{so}(3) = T_{\mathbf{I}_3} \text{SO}(3) = \{ \mathbf{R} \in \mathbb{R}^{3,3} : \mathbf{R}^\top = -\mathbf{R} \}.$$

$\dim(\mathfrak{so}(3)) = \dim(\text{SO}(3)) = 3$, with basis $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$:

$$\mathbf{L}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad \mathbf{L}_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

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Note: $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$ — the same structure as the cross product on $\mathbb{R}^3$!

The Special Unitary Group $SU(2)$

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$$SU(2) = \left\{ U \in \mathbb{C}^{2,2} : U^\dagger U = I_2, \det(U) = 1 \right\};$$

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Note: $[T_i, T_j] = \varepsilon_{ijk} T_k$ — **identical** to $\mathfrak{so}(3)$! This is the first hint of the isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

The Double Cover $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$

Construction

For $U \in \mathrm{SU}(2)$, define $\Phi(U) : \mathfrak{su}(2) \rightarrow \mathfrak{su}(2)$ by

$$[\Phi(U)](X) \stackrel{\mathrm{df}}{=} UXU^\dagger.$$

Using the basis $\beta = (T_1, T_2, T_3)$, identify $\mathfrak{su}(2) \cong \mathbb{R}^3$. Then the matrix $[\Phi(U)]_\beta^\beta$ lies in $\mathrm{SO}(3)$.

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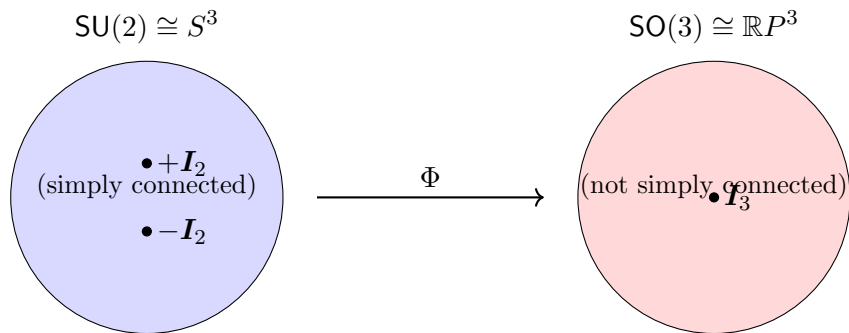
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$\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ is:

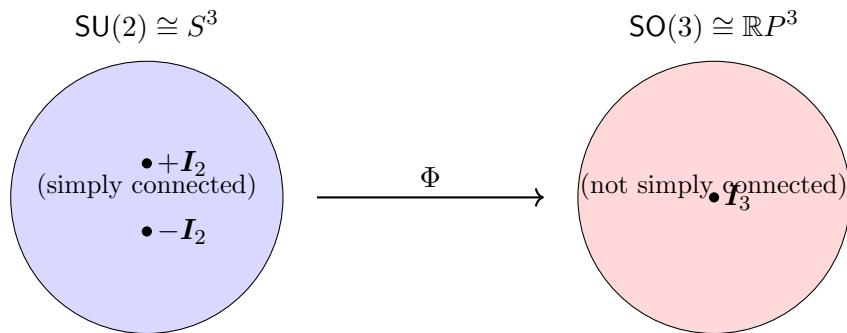
- 1 A smooth group homomorphism: $\Phi(UV) = \Phi(U) \circ \Phi(V)$.
- 2 Surjective.
- 3 Has kernel $\ker(\Phi) = \{\pm I_2\}$ (two elements).

Thus $\mathrm{SU}(2)$ is a **double cover** of $\mathrm{SO}(3)$.

The Double Cover: Geometric Picture



The Double Cover: Geometric Picture



Why two-to-one? Both U and $-U$ give the same rotation. A rotation by angle θ about axis $\hat{n}$ lifts to $\pm \cos(\theta/2)I_2 \mp \sin(\theta/2)(\hat{n} \cdot \sigma)$. Increasing θ by 2π flips the sign!

Lie Algebra Isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$

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Define $\varphi : \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$ by $\varphi(\mathbf{L}_i) \stackrel{\text{df}}{=} \mathbf{T}_i$ (and extend $\mathbb{R}$ -linearly). Then φ is a Lie algebra isomorphism.

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Proof (essence)

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Significance: Although $SU(2)$ and $SO(3)$ are different as groups (double cover vs. quotient), their Lie algebras are **identical**. Infinitesimal rotations are the same in both pictures.

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- 6 $SO(3)$ (rotations) and $SU(2)$ (unitary 2×2 matrices) are the central examples.
- 7 $SU(2)$ is a **double cover** of $SO(3)$. Their Lie algebras are **isomorphic**: $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

What Comes Next (Part II)

Representation Theory (next meeting)

- Representations of Lie algebras and Lie groups.
- Complexification $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$ and the Cartan–Weyl basis $(\mathbf{H}, \mathbf{E}, \mathbf{F})$.
- Classification of irreducible representations of $\mathfrak{su}(2)$: for each $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, a unique irrep of dimension $2j + 1$.
- Integration: every $\mathfrak{su}(2)$ -representation exponentiates to an $SU(2)$ -representation.
- Projective representations and the descent to $SO(3)$: integer $j \rightarrow$ ordinary representations, half-integer $j \rightarrow$ spinor (projective) representations.
- Physical spin operators and the 4π periodicity of spinors.

- Hall, B. C. (2015). *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction* (2nd ed.). Springer.
- Lee, J. M. (2012). *Introduction to Smooth Manifolds* (2nd ed.). Springer.
- Steinacker, H. (2024). *Lie Groups and Lie Algebras for Physicists*. Lecture notes, University of Vienna.
- Full paper with proofs: github.com/ganddoffcha/quantum-spin

Thank you!

Questions?